

coa practical

LAB 7: Design and Simulation of a 4-Stage Finite State Machine (FSM) in VHDL

Objective:

1. To design and simulate a 4-state Moore-type FSM using VHDL.
2. To verify state transitions and output sequence through waveform simulation.
3. To analyze the influence of reset and enable control signals on FSM operation.

Description:

A Finite State Machine is a sequential logic circuit that changes its internal state on each clock edge according to inputs.

FSMs are used in control units, instruction sequencing, and various digital controllers.

In this experiment, a 4-stage Moore FSM has to be designed.

The FSM has four states (S0, S1, S2, S3) and generates a unique 4-bit output pattern for each state.

Transitions occur sequentially when enable = '1', and the FSM resets to S0 when reset = '1'.

State Transition Sequence:

S0 → S1 → S2 → S3 → S0 → ...

Output Pattern:

State	Binary Code	Output (out_sig)
S0	00	0001
S1	01	0010
S2	10	0100
S3	11	1000

Procedure:

1. Start the VHDL simulator and create a new project. ✓
2. Declare an entity with inputs: clk, reset, enable; outputs: out_sig, state_id. ✓
3. Define an enumerated state_type with four states: S0, S1, S2, S3.
4. Implement the state register (synchronous with clock, asynchronous reset).
5. Write next-state logic using a case statement for cyclic transitions.
6. Implement output logic (Moore type) to assign a unique output for each state.
7. Simulate the FSM and observe state transitions and outputs.

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when others

=> next_state = S0

control → Branch

Data

structural → ok

fsm-4-state

memory

CPU

DMA

i/o device